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B. Tech Mid Term Examination
April 2022

Name of the Subject: Network Analysis and Synthesis

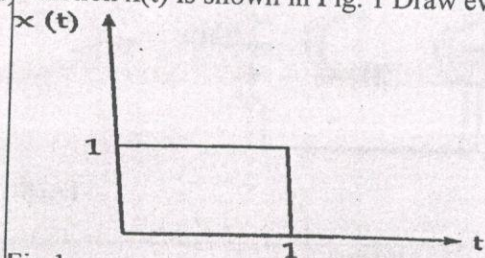
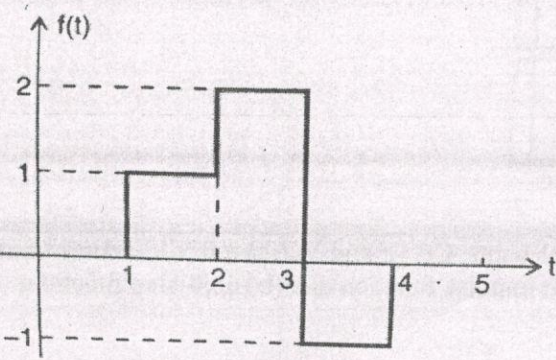
TIME 1:30 Hr.

Subject Code: ETEC-206

M.M. 30

Semester: Fourth

- . Attempt three questions. All question carry equal marks.
- Question No 1 is compulsory

Q. No.	From UNIT – I & UNIT – II	Marks
1.	a) Consider a system S whose input $x(t)$ and output $y(t)$ are related by $y(t) = t x(t)$. Determine whether or not S is linear.	(2)
	b) Sketch the following signals. (i) $t^2 [U(t-1)-U(t-3)]$ (ii) $(t-4)[U(t-1)-U(t-4)]$	(2)
	c) For the current $i(t) = 5 U(t)-3e^{-2t}$, Find $I(s)$ and hence to determine the value of $i(0^+)$ and $i(\infty)$.	(2)
	d) Function $x(t)$ is shown in Fig. 1 Draw even and odd parts of $x(t)$.	(2)
	 Fig.1	
	e) (i) Differentiate between Network and Circuit. (ii) Differentiate between transient and steady state response.	(2)
Q, N.o 2	FROM- I UNIT	
2.	a) Find the laplace transform of the waveform shown in the Fig. 2	(5)
	 Fig.2	
	b) Write the mathematical expression for the waveform shown in Fig. 3	(5)

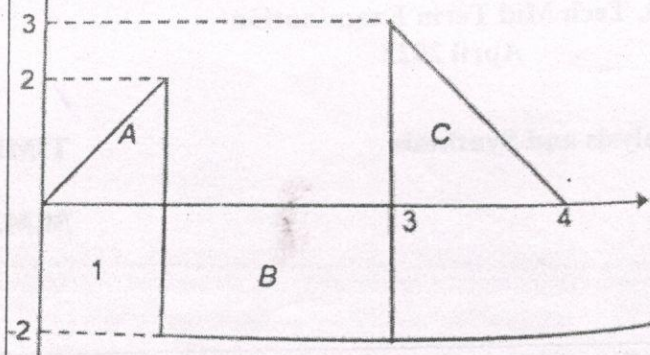


Fig. 3

Q.N.o

FROM UNIT – II

- 3 a) An exponential voltage $v(t) = 8e^{-t}$ is suddenly applied at the time $t=0$ to a series RC circuit comprising of resistor $R= 2$ ohms and capacitor $C= 0.25$ farad. Obtain the particular solution for current $i(t)$ through the circuit. Assume zero initial charge across the capacitor. (5)

- b) Obtain $i(t)$ if the pulse shown in Fig 4a is given as $v(t)$ in circuit shown in Fig 4b. Also determine the laplace transform of the pulse. (5)

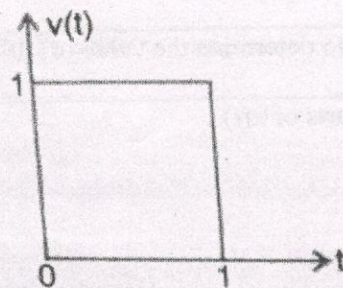


Fig.4a

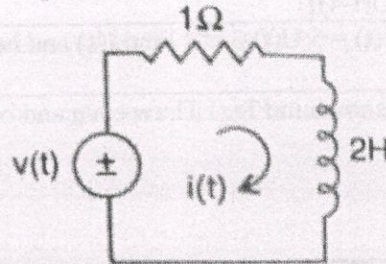


Fig.4b

Q.N.o.

From UNIT – I & UNIT - II

- 4 a) In the Fig. 5 shown, the switch S is closed. Find the time when the current from the battery reaches to 500mA. (5)

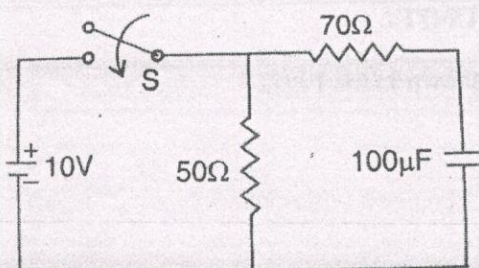


Fig. 5

- b) For the circuit shown in Fig.6 with $R=1$ ohms, $C= 1F$ and $V_c(0) = 0$ V, determine output response $v(t)$ when input $i(t)$ is (a) unit impulse function and (b) unit- step function. (5)

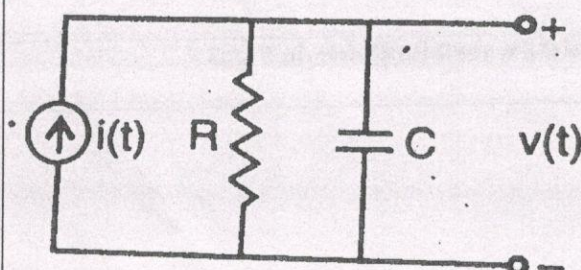


Fig. 6